

Sleep Study on the Job

An investigation into sleep qualities and occupation.

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In this report, we aim to derive potential connections between sleep behaviors, and occupation. We will be working with a dataset containing a mixture of occupations, and utilizing sleep quality, sleep duration, and stress levels, to measure correlations.

Importance and Context

We plan to study an answer to the question: “Is there a relationship between sleep quality and different occupations?”. This question is interesting to us, as we believe it is a common belief that some occupations demand more hours, or are more stressful than others may be. We wonder if this thought is based in fact, or if people, when studied, claim and behave differently in their twilight hours. There are also related questions we could seek to answer, such as “Does occupation have a notable relation to stress level?” and “Does Occupation have correlation to physical activity level?”. The question of whether occupation affects sleep has been studied before, with a mixture of results depending on the study, and individuals polled. Some look at the prevalence of short sleep in certain career fields (Oxford Academic). While others look at associations between working hours and sleep (Taylor & Francis). And some study occupational stress and sleep (NIH). We seek to evaluate a person’s claim of sleep quality first and foremost alongside their sleep duration to get a strong measurement of their occupations’ possible connection.

To answer this question, we have a data set containing occupations, sleep durations, quality, stress level, physical activity level, and even steps. We can derive information from this, by comparing each occupation and finding unique takeaways from this data set involving sleep duration/quality and career field specifically. We can use linear regression models to find correlations between sleep quality/duration and occupation in our studies, looking in particular for p-values of notable significance. Scatter plots could be useful as well to show trends and potential clustering towards, or away from high sleep quality/duration when considering different occupations. We are excited to immerse ourselves further into this data, and derive potentially notable insights.

Data and Methodology

In our research, we utilized a data set containing sleep and health/lifestyle data with 374 individual unique Person IDs. We have narrowed this information down further to exclude occupations that we deem to contain too little information to include in our model (There were too few entrants to that occupation). We set the cutoff for this at 20, meaning, we removed occupations that made up around 5% or less of the population. This helped to ensure more accuracy in our models, and brought us down to 7 Occupations out of the original 10. This still left us with a total of 363 rows of data, which we deemed sufficient for this study. Each remaining row contains multiple observations, collected at once from each individual. We specifically would like to focus on the Y-variable of sleep quality, as we noticed unexpected responses when compared to sleep duration that created notable diversity between our occupations. Sleep quality is a 1-10 **Y** measurement that we will be treating as a continuous variable in this case. We will also be comparing captured sleep quality and comparing it to sleep duration, stress level, and even physical activity. (add information on Outcome variable, observation types,)

First we can run a prediction on Occupation to give a strong visual, and next, we will move to quality of sleep.

Exploration Occupation:

Call:

```
multinom(formula = Occupation ~ Sleep.Duration + Quality.of.Sleep +  
          Stress.Level + Physical.Activity.Level, data = exploration_set)
```

Coefficients:

	(Intercept)	Sleep.Duration	Quality.of.Sleep	Stress.Level
Doctor	-68.06958	26.25839	-17.79808	1.737652
Engineer	-81.57760	25.42820	-15.30429	1.860398
Lawyer	-69.50012	21.79130	-13.78418	1.880485
Nurse	-10.57843	19.65986	-18.18884	-1.385273
Salesperson	42.40311	18.94268	-22.90091	-2.992668
Teacher	51.82075	19.20222	-22.30718	-4.743488

	Physical.Activity.Level
Doctor	0.1802392
Engineer	0.1793258
Lawyer	0.2241017
Nurse	0.3238145
Salesperson	0.2040496
Teacher	0.1017758

Std. Errors:

	(Intercept)	Sleep.Duration	Quality.of.Sleep	Stress.Level
Doctor	14.06039	5.704398	5.832040	1.0486649
Engineer	13.66953	5.700050	5.774152	0.9783598
Lawyer	12.36582	5.707371	5.663608	0.8875793
Nurse	15.35186	5.867901	6.103602	1.2497542
Salesperson	11.62755	6.405260	5.930321	0.7345807
Teacher	13.99574	6.395205	5.911389	1.0424093

	Physical.Activity.Level
Doctor	0.09573918
Engineer	0.09318041
Lawyer	0.09644472
Nurse	0.10934516
Salesperson	0.10146398
Teacher	0.11431979

Residual Deviance: 190.6752

AIC: 250.6752

Call:

```
multinom(formula = Quality.of.Sleep ~ Sleep.Duration + Occupation +
  Stress.Level + Physical.Activity.Level, data = exploration_set)
```

Coefficients:

	(Intercept)	Sleep.Duration	OccupationDoctor	OccupationEngineer
6	199.4703	-205.72644	761.6820	483.0335
7	-86.5897	101.04448	176.1374	-586.3328
8	119.7563	-129.68506	-1142.4922	-347.3975
9	-250.0201	16.30565	622.5588	332.5461

	OccupationLawyer	OccupationNurse	OccupationSalesperson	OccupationTeacher
6	553.493737	-138.4450	417.0597	670.94548766
7	-110.655442	-572.1301	-177.6027	-0.06848371
8	-346.734947	-1572.7690	-169.4891	-72.14816410
9	6.408397	465.5886	167.9355	-258.46324428

	Stress.Level	Physical.Activity.Level
6	124.51653	4.212367
7	-11.04546	-1.137882
8	-72.84385	32.789119
9	-101.10563	7.066753

Std. Errors:

	(Intercept)	Sleep.Duration	OccupationDoctor	OccupationEngineer
6	3.907974e-04	2.381316e-03	5.611631e-10	NaN

7	6.004678e+02	2.958935e+03	3.505212e+02	9.375076e+02
8	1.366993e+01	3.695826e+03	1.824155e-71	1.972800e-109
9	5.870005e+02	5.177692e+03	3.505212e+02	9.375076e+02
	OccupationLawyer	OccupationNurse	OccupationSalesperson	OccupationTeacher
6	3.743623e-122	3.905940e-04	9.635662e-22	2.478798e-15
7	1.366993e+01	1.354369e-45	1.218543e-21	2.478798e-15
8	1.366993e+01	3.090141e-12	1.708927e-104	1.957039e-27
9	0.000000e+00	3.905956e-04	0.000000e+00	0.000000e+00
	Stress.Level	Physical.Activity.Level		
6	3.124765e-03	3.515334e-02		
7	1.828352e+03	9.336591e+02		
8	6.834968e+01	8.201961e+02		
9	1.761000e+03	1.682107e+03		

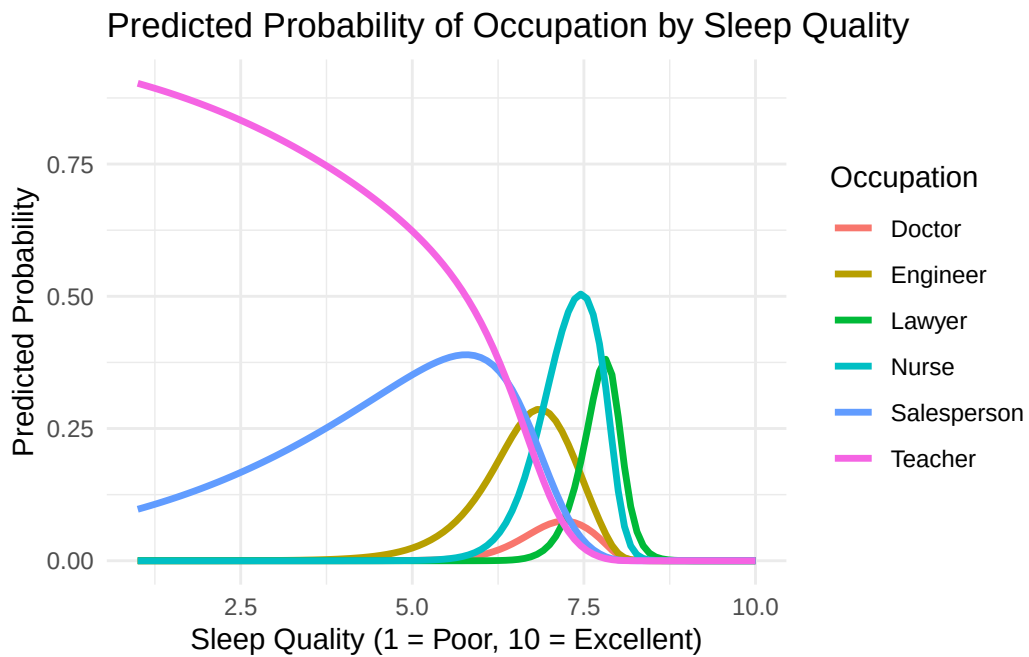
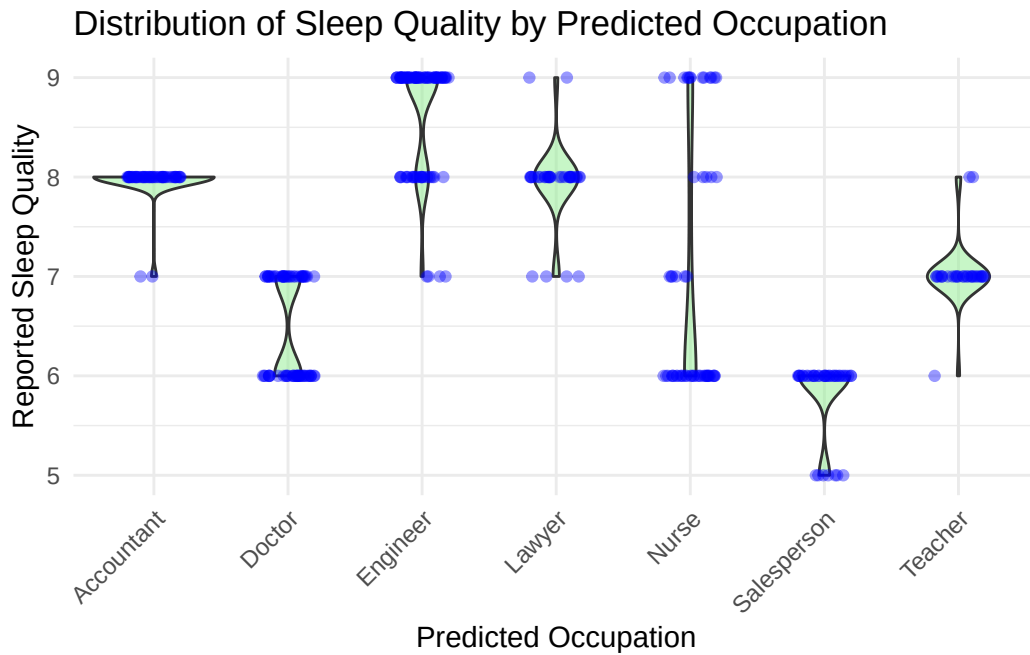
Residual Deviance: 0.0001276257
AIC: 80.00013

When running a prediction on this fitted model, we can see that this model correctly predicts Occupation our key **X** variable at a rate above average in some occupations, but shows extreme accuracy in others such as “Teacher”, “Doctor”, and “Salesperson”. We used a similar format below to predict quality of sleep, our **Y**-variable.

Next:

Filtered Data of Only Sleep Quality:

You will notice some larger standard errors on our intercept here. This is caused by variance in the population sizes of some of our occupations (Nurses were quite popular in this data set at a count of 73).



Results

In this Confirmation set

	Actual						
Predicted	Accountant	Doctor	Engineer	Lawyer	Nurse	Salesperson	Teacher
Accountant	20	0	2	9	0	0	4
Doctor	0	48	2	4	1	0	3
Engineer	0	0	24	1	17	0	0
Lawyer	4	0	10	17	8	0	0
Nurse	2	2	4	2	22	0	0
Salesperson	0	0	1	0	4	23	3
Teacher	0	0	2	0	0	0	18

Confirmation set 2

	Actual									
Predicted	1_QoS	2_QoS	3_QoS	4_QoS	5_QoS	6_QoS	7_QoS	8_QoS	9_QoS	10_QoS
1_QoS	0	0	0	0	0	0	0	0	0	0
2_QoS	0	0	0	0	0	0	0	0	0	0
3_QoS	0	0	0	0	0	0	0	0	0	0
4_QoS	0	0	0	0	0	0	0	0	0	0
5_QoS	0	0	0	0	4	3	0	0	0	0
6_QoS	0	0	0	0	0	67	0	0	0	0
7_QoS	0	0	0	0	1	2	45	0	0	0
8_QoS	0	0	0	0	0	0	7	72	0	0
9_QoS	0	0	0	0	0	0	2	3	50	0
10_QoS	0	0	0	0	0	0	0	0	0	0

Call:

```
multinom(formula = Quality.of.Sleep ~ Sleep.Duration + Occupation +
  Stress.Level + Physical.Activity.Level, data = exploration_set2)
```

Coefficients:

```
(Intercept) Sleep.Duration OccupationDoctor OccupationEngineer
6      8240.042      -3207.584      11825.72      -3283.633
7     -4600.831       3743.322      15345.05      16511.828
8     10833.813       2443.137     -29070.62       2077.434
9    -13807.139       8719.416      18741.80       6365.036
OccupationLawyer OccupationNurse OccupationSalesperson OccupationTeacher
6       3766.646      -5578.711      11172.598      -13383.778
7       8112.222       6959.132      -7329.157       6580.608
8      -11847.613     -14000.328     -9060.170     -11252.438
9      -1127.851      -7062.172      10496.978     -9320.419
Stress.Level Physical.Activity.Level
6      -1418.296       392.1441
```

7	-4791.981	133.1879
8	-5939.215	464.5783
9	-18015.675	737.3199

Std. Errors:

	(Intercept)	Sleep.Duration	OccupationDoctor	OccupationEngineer
6	NaN	0.000000e+00	0	0
7	1.663260e-04	1.230812e-03	0	0
8	1.663260e-04	1.230812e-03	0	0
9	4.340768e-80	3.429207e-79	0	0

	OccupationLawyer	OccupationNurse	OccupationSalesperson	OccupationTeacher
6	0.000000e+00	NaN	0	0
7	1.663260e-04	2.85742e-12	0	0
8	1.663260e-04	2.85742e-12	0	0
9	4.340768e-80	0.00000e+00	0	0

	Stress.Level	Physical.Activity.Level
6	NaN	NaN
7	8.316300e-04	9.979560e-03
8	8.316300e-04	9.979560e-03
9	2.170384e-79	3.906691e-78

Residual Deviance: 3.819085

AIC: 83.81909

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Occupation	6	194.8	32.46	41.65	<2e-16 ***
Residuals	356	277.5	0.78		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Tukey multiple comparisons of means
95% family-wise confidence level

Fit: aov(formula = Quality.of.Sleep ~ Occupation, data = dat)

\$Occupation

	diff	lwr	upr	p adj
Doctor-Accountant	-1.244004568	-1.77482611	-0.713183023	0.0000000
Engineer-Accountant	0.520806521	-0.02143849	1.063051530	0.0690249
Lawyer-Accountant	0.001725129	-0.57365724	0.577107501	1.0000000
Nurse-Accountant	-0.522028878	-1.05035338	0.006295622	0.0552002
Salesperson-Accountant	-1.891891892	-2.52388944	-1.259894342	0.0000000

Teacher-Accountant	-0.916891892	-1.51403882	-0.319744960	0.0001469
Engineer-Doctor	1.764811089	1.31168439	2.217937786	0.0000000
Lawyer-Doctor	1.245729697	0.75342996	1.738029431	0.0000000
Nurse-Doctor	0.721975690	0.28560322	1.158348158	0.0000291
Salesperson-Doctor	-0.647887324	-1.20530553	-0.090469118	0.0112477
Teacher-Doctor	0.327112676	-0.19045723	0.844682586	0.4989307
Lawyer-Engineer	-0.519081391	-1.02367743	-0.014485350	0.0391550
Nurse-Engineer	-1.042835399	-1.49303432	-0.592636478	0.0000000
Salesperson-Engineer	-2.412698413	-2.98100572	-1.844391107	0.0000000
Teacher-Engineer	-1.437698413	-1.96697787	-0.908418951	0.0000000
Nurse-Lawyer	-0.523754008	-1.01336027	-0.034147745	0.0272005
Salesperson-Lawyer	-1.893617021	-2.49362404	-1.293610007	0.0000000
Teacher-Lawyer	-0.918617021	-1.48179725	-0.355436790	0.0000403
Salesperson-Nurse	-1.369863014	-1.92490384	-0.814822185	0.0000000
Teacher-Nurse	-0.394863014	-0.90987163	0.120145604	0.2598364
Teacher-Salesperson	0.975000000	0.35409099	1.595909012	0.0000925

[1] NA

Evaluate accuracy or confusion matrix table(Predicted = confirmation_set*Predicted*, Actual = confirmation_setOccupation)

Discussion

Final Report

Your final report should document your analysis, communicating your findings in a way that is technically precise, clear, and persuasive.

Page limits:

- Main report: 4 pages

- Appendix: 1 additional page

You must meet these page limits using standard `pdf_document` output in a `.Rmd` file, or in a `documentclass: scrartcl` in a `.qmd` file. (This assignment prompt has been generated as a `scrartcl`).

The one-page appendix is intended for extra information that will help your instructor assess your model building process. Please include the following elements:

1. **A Link to your Data Source.** If you used specialized code to access your data, please include that here. Please make sure your instructor has the ability to access the data.
2. **A List of Model Specifications you Tried.** We are interested in seeing how you arrived at your final model. In just a sentence, please provide a reason or something that you learned from each specification.
3. **A Residuals-vs-Fitted-values Plot.** Please generate this plot for your final model that includes variable transformations. Your instructor will use this plot to assess how well you have captured the shape of the relationship between your X and your Y variable. For example, if there is a clear parabolic pattern in your residuals-vs-fitted plot, that is a signal that you should have included a square term.

Evaluation Criteria

We present the following criteria to guide you to a professional-quality report. Moreover, these criteria are also the ones we will use to grade your report. The descriptions below are copied directly from our grading rubric.

1. Intro

An introduction that is scored in the top level has very successfully made the case for the study. It will have explained why the topic is interesting, and provided compelling reasons to care about not just the general area, but rather about every concept in the research question and the statistical results to be generated. The introduction will be engaging from the very first sentence, and create a logical story that leads the reader step-by-step to the research question. After reading the introduction, no part of the research question will appear arbitrary or unexpected, instead flowing naturally from the background provided.

2. Description of the Data Source

A report that is scored in the top level will describe the provenance of the data; the audience will know the source of the data, the method used to collect the data, the units of observation of the data, and important features of the data that are useful for judging the analysis.

3. Data Wrangling

A report that is scored in the top level on data wrangling will have succeeded to produce a modern, legible data pipeline from raw data to data for analysis. Because there are many pieces of data that are being marshaled, the reports that score in the top level will have refactored different steps in the data handling into separate files, functions, or other units. It should be clear what, and how, any additional features are derived from this data. The analysis avoid defining additional data frames when a single source of truth data would suffice.

4. Operationalization

A report that is scored in the top level on operationalization will have precisely articulated and justified the decisions leading to the variables used in the analysis. The reader will be left with a clear understanding of the concepts in the research question, and how they relate to the operational definitions, including any major gaps that may impact the interpretation of results. You should also list how many observations you remove and for what reasons. When there is more than one reasonable way to operationalize a concept, the report will explain the alternatives and provide reasons for the one that was selected. Note that there is often more than one way to operationalize a concept; we are less interested in whether you make the best possible choice, and more in how well you defend your decisions.

5. One or More Data Visualization(s)

This is a task of description, and one of the most effective ways to describe relationships between variables is through visualization.

You are required to include at least one plot in your main report that highlights the relationship between your Y and your X variables. In your text, you should provide your interpretation of what this plot means, and link to the plot. For example, we might write that Figure 1 displays 300 observations from a synthetic model of Earnings and Age, showing an increasing trend though the relationship is less positive for higher ages.

Include a visual representation of your final model predictions on the plot. A report that is scored in the top level will have plots that effectively transmit information, engage the reader's interest, maximize usability, and follow best practices of data visualization. Titles and labels will be informative and written in plain English, avoiding variable names or other artifacts of R code. Plots will have a good ratio of information to space or information to ink; a large or complicated plot will not be used when simple plot or table would show the same information more directly. Axis limits will be chosen to minimize visual distortion and avoid misleading the viewer. Plots will be free of visual artifacts created by binning. Colors and line types will be chosen to reinforce the meanings of variable levels and with thought given to accessibility for the visually-impaired. Your report will be free of "output dumps" - code output that has

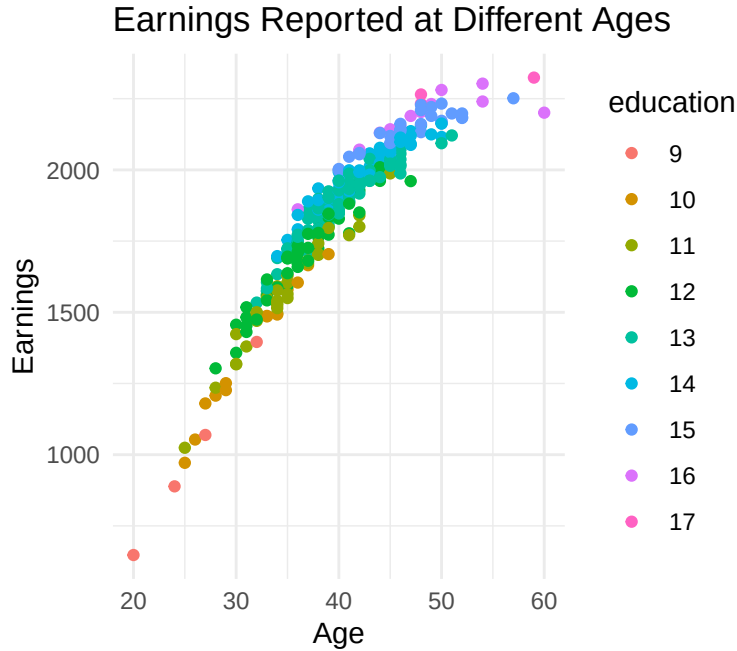


Figure 1: Synthetic data showing a hypothetical relationship between Earnings, Age, and Education. An increasing trend can be seen with decreasing slope for higher ages.

not been formatted for human-readability. Every single plot and table in the report will be discussed in the narrative of the report.

6. Model Specification

A report that is scored in the top level will have chosen a set of regression models that strongly support the goal of the study. Variables transformations will be chosen to inform the reader of the shape of the joint distribution, and will be human-understandable. A reason will be provided for the chosen variable transformations. Ordinal variables will not be treated as metric. All model specifications will be displayed in a regression table, using a package like [stargazer](#) to format your output. Displayed standard errors will be correctly chosen.

```
#“{r report models, message = FALSE, warning = FALSE, results = ‘asis’} ##| results:
asis ##| warning: FALSE ##| message: FALSE ##| tbl-cap: Regression Results. ##| #|
tbl-pos: t #stargazer( # model_one, # model_two, # model_three, # header = FALSE,
# se = list( ## sqrt(diag(vcovHC(model_one))), # sqrt(diag(vcovHC(model_two))), #
sqrt(diag(vcovHC(model_three))) # ), # type = ifelse(knitr::is_latex_output(), “latex”,
“html”), # covariate.labels = c(“Age”, “
textAge2”, “Education”, “Intercept”), # digits = 2, # omit.stat = c(‘ser’, ‘f’), # star.cutoffs
= c(0.05, 0.01, 0.001) #)
```

7. Model Assumptions

A report that scores in the top-level has provided an thorough and precise assessment of the assumptions supporting the regression. The list of assumptions is appropriate given the sample size and modeling goals. Each assumption is evaluated fairly, and discussed defensively - without overstating how credible the assumption is or rendering a final up-or-down judgement on the assumption. The report will not miss any important violations of any assumption. Where possible, the report discusses the consequences for the analysis of a violated assumption.

8. Model Results and Interpretation

A report that scores in the top level will correctly interpret statistical significance, clearly interpret practical significance, and comment on the broader implications of the results. It may want to include statistical tests besides the standard t-tests for regression coefficients. When discussing practical significance, comment on both the direction and magnitude of your coefficients, placing them in context so the reader can understand if they are important. To help the reader understand your fitted model, you may want to describe hypothetical datapoints (e.g. a hypothetical person with 1 cat is predicted to spend \$2400 on pet care. That rises to \$3200 for a hypothetical person with 2 cats...).

9. Overall Effect

A report that scores in the top level will have met expectations for professionalism in data-based writing, reasoning and argument for this point in the course. It can be presented, as is, to another student in the course, and that student could read, interpret and take away the aims, intents, and conclusions of the report.

References